

Input

Conformer Ensemble
n=50 per mol

Morgan FP
2048-bit, r=2

Features

Mean μ
D-dim

Cov Σ
5 invariants

FP
2048-dim

Models

DKO Gated
 μ/Σ fusion

Attention
Conformer weights

Hybrid XGBoost
 $\text{FP} + \mu + \Sigma$

Property Prediction
RMSE evaluation

Key Finding

FP beats all neural methods alone

BUT

improves ESOL by 9.9%
 $\text{FP} + \mu + \Sigma$
FreeSolv by 3.9%
QM9-HOMO by 4.2%